# MARK SCHEME · TEACHER COPY

10BR · Chapter 1: Data Representation

**Cambridge IGCSE Computer Science 0478**

Paper 1  Computer Systems

## Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

For teacher / admin use only. Do not give to students before they have attempted the paper.

Mark scheme extracts © Cambridge University Press & Assessment.

## Question 1 · May/June 2023 Paper 12, original Q3 (a),(b) [10 marks]

| Question | Answer | Marks |
|---|---|---|
| 1(a) | Any **three** from:<br>• A character set is used<br>• … such as Unicode/ASCII<br>• Each character has a **unique binary** value | 3 |
| 1(b)(i) ) | • It reduces the **file** size | 1 |
| 1(b)(ii) | Any **four** from:<br>• A compression algorithm is used<br>• … such as RLE/run length encoding<br>• **Repeating** words/characters/phrases are identified // Patterns are identified<br>• … and indexed<br>• … with number of occurrences<br>• … with their position | 4 |
| 1(b)(iii) | Any **two** from:<br>e.g.<br>• To save **storage** space<br>• To make it quicker to transmit<br>• To make it small enough to attach to an email<br>• To reduce the bandwidth needed to transmit | 2 |

## Question 2 · Oct/Nov 2023 Paper 13, original Q2 (a),(b),(c),(d),(e) [13 marks]

| Question | Answer | Marks |
|---|---|---|
| 2(a) | Any **two** from:<br>– It has a base of 2<br>– It only uses two **values**<br>– … that are 1 and 0 | 2 |

| | | |
|---|---|---|
| 2(b) | – (0000)1110<br>– (00)111011<br>– 11101010 | **3** |
| 2(c) | – 9<br>– 1A<br>– 41 | **3** |
| 2(d) | **One** mark for suitable working method e.g. conversion to binary<br>**One** mark for correct answer<br><br>– 01111011 | **2** |
| 2(e) | **One** mark for each correct nibble (max 2)<br>**One** mark for correct working e.g. correct carries<br>1 1 1<br>0 0 1 1 0 0 1 1<br>+ 0 1 1 1 1 0 0 0<br>1 0 1 0 1 0 1 1 | **3** |

**Question 3 · Feb/March 2024 Paper 12, original Q8 (a),(b),(c) [7 marks]**

| Question | Answer | Marks |
|---|---|---|
| 3(a)(i) | EC | 1 |
| 3(a)(ii) | Any **one** from:<br>• Easier for humans to read/remember<br>• Shorter for humans to enter<br>• Less likely for humans to make mistakes<br>• Easier for humans to spot errors/debug<br>• Takes up less space **onscreen** | 1 |

| | | |
|---|---|---|
| 3(b)(i) | **One** mark for working, **one** mark for answer<br>• e.g. showing flip and add 1<br>10110111 = 01001001<br>• –73 | 2 |
| 3(b)(ii) | 00101101 | 1 |
| 3(c) | **One** mark each<br>• Divide<br>• ...by 16 / $2^4$ | 2 |

## Question 4 · May/June 2024 Paper 12, original Q1 (a),(b),(c),(d) [6 marks]

| Question | Answer | Marks |
|---|---|---|
| 4(a) i) | (1) byte | 1 |
| 4(b) | 8192 | 1 |
| 4(c) | (1) Tebibyte // TiB | 1 |
| 4(d) | **One** mark for each correct stage of working (**max** 2):<br>• 512 × 512<br>• 262 144 * 2 // multiplied by 16 and divided by 8<br>• 524 288/1024<br><br>**One** mark for the correct answer:<br><br>512 (KiB) | 3 |

## Question 5 · Oct/Nov 2024 Paper 11, original Q3 (a),(b),(c),(d) [11 marks]

| Question | Answer | Marks |
|---|---|---|
| 5(a) i) | Unicode | 1 |
| 5(b)(i) | • (0)1000001<br>• (0)1101101 | 2 |

**Total: 75 marks**

| | | |
|---|---|---|
| 5(b)(ii) | • 41<br>• 6D | **2** |
| 5(c)(i) | 121 | **1** |
| 5(c)(ii) | 79 | **1** |
| 5(c)(iii) | 00011110 | **1** |
| 5(d) | **One** mark for correct working, for example: carries<br>**One** mark for each correct nibble.<br><br>1 1 1 1<br>0 1 0 1 0 1 0 0<br>0 1 1 1 0 1 0 0<br>1 1 0 0 1 0 0 0 | **3** |

**Question 6 · Oct/Nov 2024 Paper 11, original Q6 (c) [1 marks]**

| Question | Answer | Marks |
|---|---|---|
| 6(a) | Any **one** from:<br><br>• Convert to hexadecimal<br>• Convert to denary<br>• A character set can be used | **1** |

**Question 7 · Feb/March 2025 Paper 12, original Q1 (a),(b),(c) [11 marks]**

| Question | Answer | Marks |
|---|---|---|

| | | |
|---|---|---|
| 7(a) | 1 mark for each correct item (in bold)<br><br>The binary number system is base **2**. The smallest denary number that can be represented an 8-bit binary number is **0**. The largest denary number that can be represented an 8-bit binary number is **255**.<br>The hexadecimal number system is base **16**. Each hexadecimal digit is equivalent to **4** bits. The numbers 1 to 9 are used and the number 10 is represented by **A**. The hexadecimal number system continues up to the number 15, which is represented by **F**. | **7** |
| 7(b) | 1 mark for correct working, 1 mark for each correct nibble<br>For example:<br><br>1 0 0 1 1 0 1 1<br>+ 0 0 0 1 0 0 1 1<br>1 0 1 0 1 1 1 0<br>1 1 1 | **3** |
| 7(c) | C | **1** |

**Total: 75 marks**

## Question 8 · Oct/Nov 2025 Paper 12, original Q2 (a),(b) [7 marks]

| Question | Answer | Marks |
|---|---|---|
| 8(a) | One mark for each correct nibble (2)<br>One mark for a correct method of working for example 4 or 5 correct carries<br>One mark for showing an overflow error<br><br>(1) 111 1<br>11110101<br>+ 00111001<br>1 0010 1110 | 4 |
| 8(b) | One mark for showing the binary number after the shift<br>One mark for valid method of working to convert to denary<br>One mark for correct denary number<br><br>• 00011110<br>• 16 + 8 + 4 + 2<br>• 30 | 3 |

## Question 9 · May/June 2026 Paper 11, original Q3 (a),(b),(c),(d),(e),(f) [9 marks]

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 9(a) | **One** from:<br><br>• Denary // decimal<br>• Two's complement | 1 | Mark the first answer given.<br><br>Accept other system outside syllabus e.g. octal, BCD<br><br>BOD Base-10 |
| 9(b) | 0010 1010 | 1 | Mark the first answer given.<br><br>Correct number of leading zeros must be present<br><br>If no answer is written in the answer space an obvious correct answer can be awarded in the working space. |

| | | | |
|---|---|---|---|
| 9(c) | 0001 0000 0001 | **1** | Mark the first answer given.<br><br>Correct number of leading zeros must be present<br><br>If no answer is written in the answer space an obvious correct answer can be awarded in the working space. |
| 9(d) | 71 | **1** | Mark the first answer given.<br><br>If no answer is written in the answer space an obvious correct answer can be awarded in the working space. |
| 9(e) | 2D8 | **1** | Mark the first answer given.<br><br>If no answer is written in the answer space an obvious correct answer can be awarded in the working space. |
| 9(f) | **One** mark for each correct nibble (MAX 2)<br>**One** mark for the three correct carries as working<br>**One** mark for one overflow bit<br><br>For example:<br><br>1 1 1<br>1 0 0 1 1 1 0 1<br>+ 1 1 1 1 0 0 0 0<br><br>1 1 0 0 0 1 1 0 1 | **4** | Carries can be shown above/below. Award where seen as long as for the correct columns.<br><br>BOD 4 carries shown if the overflow is shown as a carry first, then as overflow.<br><br>Ignore the crossing out of carries. Award the number of carries seen. |